



Functional Specification Document

Procurement Tracking Portal (PTP)

Contents

Document Signoff.....3

Revision History.....3

1.INTRODUCTION.....4

 1.1 Summary4

 1.2 Project Overview.....4

 1.3 Project Objectives4

 1.4 BRD Summary4

2.SCOPE OF WORK6

 2.1 Key Features6

3.TECHNICAL ARCHITECTURE8

 3.1 System Overview8

 3.2 Proposed Tech Stack8

 3.3 System Diagram.....9

4.FUNCTIONAL DESIGN10

 4.1 Purchase Request10

 4.2 Workflow.....12

 4.3 Workflow Mapping14

 14

 4.4 Dashboard14

5.DATABASE DESIGN16

 5.1 Entity Relationship Diagram (ERD)16

 5.2 Key Tables and Relationships17

6.PROJECT PLAN18

 6.1 Timeline18

Document Signoff

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Revision History

Change between document issues is cumulative. The latest document contains all the changes made in earlier issues.

Version	Status	Author	Date	Revision Summary	Reviewed By	Date
1.0	Draft	Sami Ullah	2025-04-10	Initial Draft	M. Azhar Iqbal	2025-04-15

1.INTRODUCTION

1.1 Summary

The proposed Procurement Tracking Portal (PTP) is a web-based solution designed to streamline and automate the procurement tracking process. The system aims to enhance tracking efficiency, transparency, and accountability in managing purchase requests, approvals, and procurement workflows. By incorporating user management, document uploads, hierarchical approvals, task tracking, and analytics, this portal will provide a centralized platform to optimize procurement operations and improve decision-making.

This proposal outlines the project's objectives, scope, features, technical architecture, timeline, and resource requirements to ensure successful implementation, based on discussions held on April 03, 2025, with IT attendees (Azhar Iqbal, Samiullah, Umair) and Procurement attendees (Faisal Rafiq, Nauman Asim).

1.2 Project Overview

The Procurement Tracking Portal (PTP) is a web-based solution designed to streamline departmental purchase request management, workflow tracking, and procurement execution. It aims to enhance efficiency, ensure compliance with approval hierarchies, and provide actionable insights through a centralized dashboard.

1.3 Project Objectives

- To develop a secure, scalable, and user-friendly procurement tracking system.
- To automate purchase request creation, approval workflows, and procurement execution.
- To provide real-time visibility into procurement statuses and task analytics via a dashboard.
- To reduce manual errors and processing time in procurement operations.
- To ensure compliance with organizational approval hierarchies and policies.

1.4 BRD Summary

- The respective departments will initiate purchase requests.
- Each request will follow the predefined procurement approval hierarchy.
- Once procurement review is completed, the final approval request will appear in the workflow mapping screen.

- The procurement team will create the workflow, after which it will proceed through the approval hierarchy.
- Approved workflows will be mapped to the corresponding approved department requests.
- The procurement team will manually update the workflow task status, which will be reflected on the client's purchase request interface.

2.SCOPE OF WORK

The Procurement Tracking Portal will include the following key features:

2.1 Key Features

1. User Management

- Role-based access control (e.g., Admin, Requester, Approver, Procurement Officer).
- User authentication and authorization (e.g., JWT, RBAC).
- Profile management and audit logging for user activities.

2. Purchase Request Creation with Document Uploads and Approval

- Form-based request submission with fields for request title, department, and documents.
- Support for uploading supporting documents (PDF, Word, Images; max 10MB).
- Multi-step approval process with notifications (email/SMS) to procurement team and request owner.

3. Purchase Request Procurement Tracking

- Real-time status updates for each request (e.g., Submitted, Approved, In Progress, Completed).
- History log of all actions taken on a request, editable only if rejected.

4. Approvals Hierarchy

- Configurable approval levels based on organizational structure (e.g., Manager → Director → Procurement).
- Dynamic routing of requests to appropriate approvers with rejection notifications.

5. Workflow Creation and Approval

- Customizable workflows for different procurement types (e.g., Vendor Selection, Equipment Purchase).
- Workflow templates with pre-defined tasks, submitted for approval before mapping.

6. Request and Workflow Mapping

- Linkage between requests and approved workflows (1:1 mapping, reusable workflows).
- Validation to ensure requests follow the assigned workflow.

7. Procurement Execution with Task Updates

- Task assignment to procurement team members with planned and actual dates.
- Progress tracking with status updates (e.g., Pending, Completed) and efficient metrics.

8. Dashboard for Task Analytics

- Visual representation of procurement metrics (e.g., pending requests, completed tasks, efficiency).
- Filters for date range, department, and workflow type; exportable reports in PDF/Excel formats.

3. TECHNICAL ARCHITECTURE

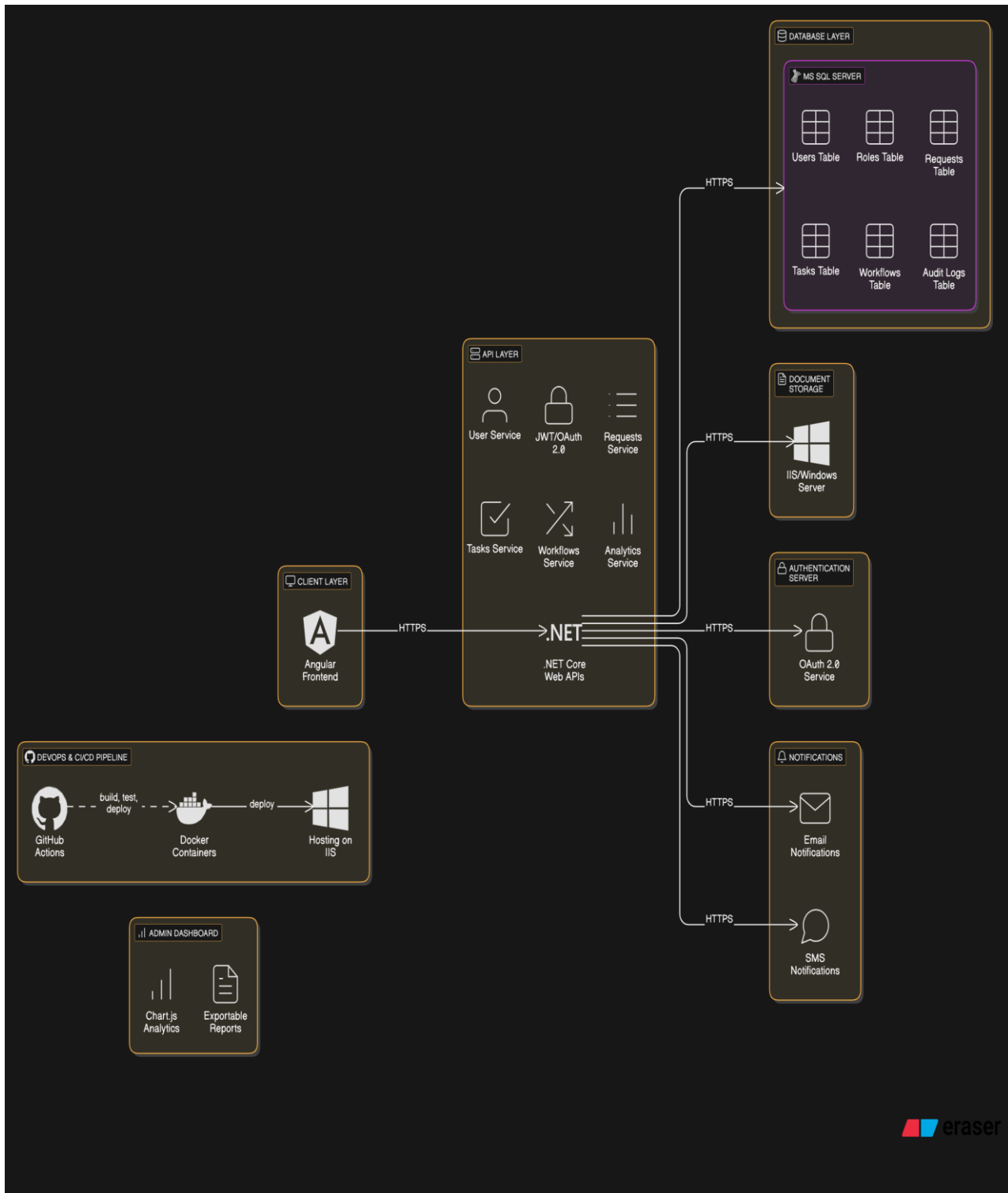
3.1 System Overview

The portal will be a web-based application built using a modern tech stack to ensure scalability, security, and maintainability.

3.2 Proposed Tech Stack

- **Frontend:** Angular, RXJS, Charts
- **Backend:** .Net(API development), RESTful architecture.
- **Database:** MSSQL (relational database for structured data), IIS(document storage).
- **Authentication:** JWT, RBAC for secure user access.
- **Hosting:** IIS
- **DevOps:** Docker (containerization), CI/CD pipeline with GitHub Actions.

3.3 System Diagram



4.FUNCTIONAL DESIGN

4.1 Purchase Request

Description: Represents the initial request raised by a department, including a document list.

- **Table:** PurchaseRequest + Document
- **Action:** Inserts into PurchaseRequest and multiple rows into Document.

Business Rules:

- PDF, Word, and Image type documents will be uploaded with max 10MB size.
- The system will ensure that all mandatory documents are uploaded before a request can be submitted.
- The system will notify the procurement department when a purchase request is generated.
- The purchase request will be editable when request is rejected.
- Approval hierarchy will be implemented on that request
- When request will be rejected by the procurement system notify to request owner

Purchase Request screen-1

Project Title:

Budget:

Document Type:

Upload Document:

+ Add

Request List

save

Department	Project Name	Documents count	Request Status	Actions
IT	SSP	3	Pending	Track Procurment,View,Request History

Track Procurement screen-2

Department :	_____	Project Title :	_____
Workflow :	_____	Total Days :	_____
Start-Date :	_____	Expected End Date:	_____
Task-1			
Title:	Document Review	Status	Running
Task-2			
Title:	Financial Assessment	Status	Pending

1. PurchaseRequest

- **Table:** AcquisitionRequests
- **Columns:**
 - RequestID (PK, INT, AUTO_INCREMENT): Unique identifier.
 - DepartmentID (FK, INT): Links to the requesting department.
 - RequestTitle (VARCHAR(100)): Name of the acquisition task.
 - Status (VARCHAR(20)): State (e.g., "Pending," "Approved").
 - CreatedDate (DATETIME): Timestamp of creation.
 - ModifiedDate (DATETIME): Timestamp of creation.

2. Document

- **Table:** AcquisitionRequestDocuments
- **Columns:**
 - DocumentID (PK, INT, AUTO_INCREMENT): Unique identifier.
 - RequestID (FK, INT): Links to the acquisition request.
 - DocumentType (VARCHAR(50)): Type (e.g., "Invoice," "PO").
 - DocumentName (VARCHAR(100)): Name (e.g., "Invoice_001.pdf").
 - DocumentAttachment (BLOB or VARCHAR(255)): File or file path/URL.

4.2 Workflow

Description: Represents the workflow created for an approved acquisition request.

- **Table: Workflow + Task**
- **Action:** Inserts into Workflow and multiple rows into Task.

BR: only approved workflow will be mapped with purchase request

Workflow Template

Workflow Type : _____ Workflow Title: _____ Select Procurement Type: _____

Procurement-owner: _____ Total Days: _____

Add-Task-1

Task Title: _____ Total Days : _____

Add-Activity-1

Activity Title: _____ Total Days : _____

Add-Activity-2

Activity Title: _____ Total Days : _____

Template List

Workflow Title	Workflow Type	Procurement Type	Total Days	Actions
Workflow	Program	Tender	62	View, Update

save

1. Workflow

- **Description:** Represents a standalone workflow, created independently.
- **Table:** Workflow
- **Columns:**
 - WorkflowID (PK, INT, AUTO_INCREMENT): Unique identifier.
 - Title (VARCHAR(100)): Workflow name (e.g., "Vendor Selection").
 - TotalDays (INT): Estimated duration in days.
 - Status (VARCHAR(20)): State (e.g., "Draft," "Approved").

- CreatedDate (DATETIME): Timestamp of creation.

2. Task

- **Description:** Individual tasks within workflow.
- **Table:** Task
- **Columns:**
 - TaskID (PK, INT, AUTO_INCREMENT): Unique identifier.
 - WorkflowID (FK, INT): Links to the parent workflow.
 - TaskTitle (VARCHAR(100)): Name (e.g., "Contact Vendor").
 - TaskDays (INT): Planned duration (read-only in procurement).
 - StartDate (DATE, NULLABLE): Actual start date.
 - EndDate (DATE, NULLABLE): Actual end date.
 - Status (VARCHAR(20)): State (e.g., "Pending," "Completed").
 - EfficiencyDays (INT, NULLABLE): Calculated as EndDate - StartDate - TaskDays.

Flow: Workflow submitted → enters approval hierarchy → approved → attached to task.

Screen:

3→ Start Procurement

Description: Maps acquisition requests to workflows (separate table as requested).

- **Table:** RequestWorkflowMapping
- **Action:** Updates Task rows with StartDate, EndDate, Status, EfficiencyDays.

BR:

4.3 Workflow Mapping

Map Workflow

Select Request: _____ Select Workflow: _____ Total Count: _____ Pending Days: _____
Mark Status : _____ Reason: _____

Add-Task-1

Task Title: _____
Total Days : _____ Task Status : _____ Start Date: _____ End Date: _____

Add-Activity-1

Activity Title: _____
Total Days : _____ Activity Status : _____

Mapped List

save

Department	Project	Workflow Title	Total Days	Pending Days	Efficient Days	Start Date	End Date	Status	Actions
Workflow	Program	Tender	20	15	5	08-04-2025	23-04-2025	Completed	View,Update

Columns:

- MappingID (PK, INT, AUTO_INCREMENT): Unique identifier for the mapping. Efficient
- RequestID (FK, INT): Links to the acquisition request.
- WorkflowID (FK, INT): Links to the assigned workflow.
- AssignmentDate (DATETIME): Timestamp of mapping.
- MappingStatus (VARCHAR(20)): State (e.g., "Assigned," "Completed").

Notes: Allows one request to map to one workflow (1:1), but workflows can be reused across requests if needed.

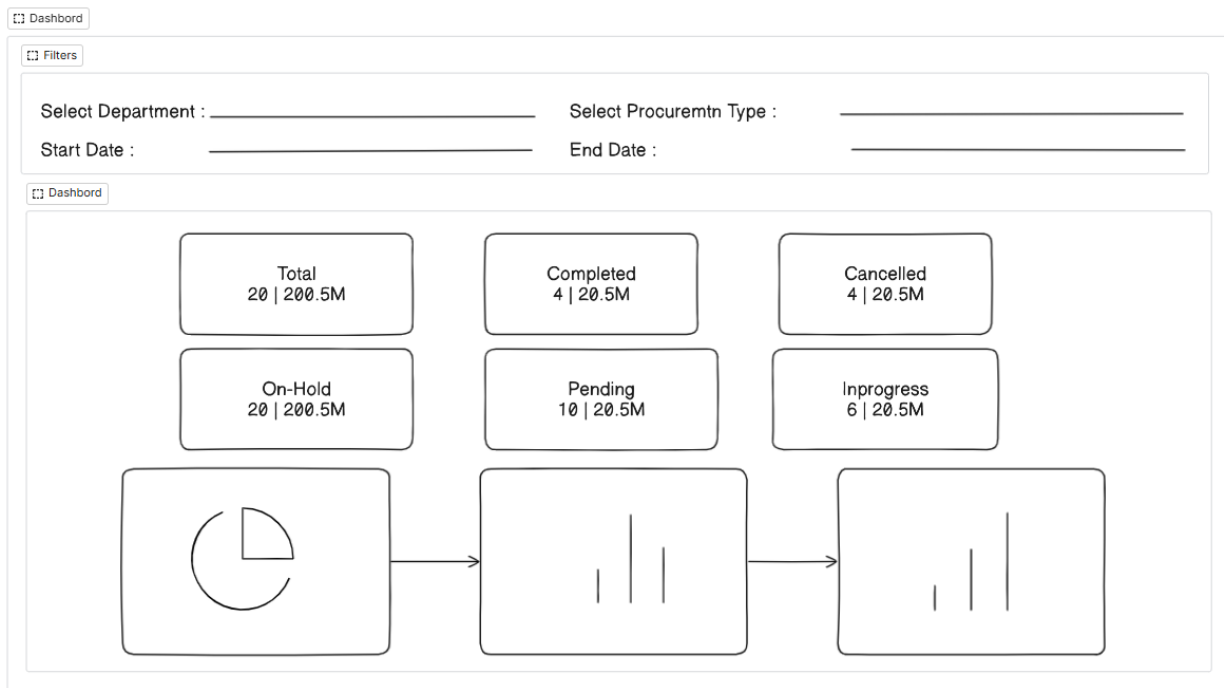
4.4 Dashboard

Description: Aggregates data for dashboard display.

- **Table/View:** DashboardData

- **Entities:** Aggregates from AcquisitionRequest, Workflow, Task, Department.
- **Fields:** Filters (WorkflowType, Department, StartDate, EndDate).
- **Metrics** (Pending/Completed/In-Progress Count and Value).
- **Action:** Queries data with JOINS and GROUP BY clauses.

BR:



Columns:

- RequestID (FK, INT): Links to acquisition request.
- WorkflowID (FK, INT): Links to workflow.
- DepartmentID (FK, INT): Links to department.
- Status (VARCHAR(20)): Request/mapping status.
- Value (DECIMAL(15,2)): Monetary value (if applicable).
- StartDate (DATE): Workflow/task start date.
- EndDate (DATE): Workflow/task end date.

5.DATABASE DESIGN

5.1 Entity Relationship Diagram (ERD)



5.2 Key Tables and Relationships

- **Department 1:N AcquisitionRequest**
 - AcquisitionRequest.DepartmentID references Department.DepartmentID.
- **AcquisitionRequest 1:N Document**
 - Document.RequestID references AcquisitionRequest.RequestID.
- **Workflow 1:N Task**
 - Task.WorkflowID references Workflow.WorkflowID.
- **Workflow 1:N Approval**
 - Approval.WorkflowID references Workflow.WorkflowID.
- **AcquisitionRequest N:1 RequestWorkflowMapping**
 - RequestWorkflowMapping.RequestID references AcquisitionRequest.RequestID.
- **Workflow N:1 RequestWorkflowMapping**
 - RequestWorkflowMapping.WorkflowID references Workflow.WorkflowID.
- **Department 1:N User**
 - User.DepartmentID references Department.DepartmentID.
- **User 1:N Approval**
 - Approval.ApproverID references User.UserID.

6.PROJECT PLAN

6.1 Timeline

The project will be executed in phases over an estimated **8 weeks (50 days)**, starting April 07, 2025, based on resource availability and feedback cycles.

Phase	Tasks	Workdays
1. Requirements Analysis	Stakeholder interviews, finalize scope (07-04-2025 to 11-04-2025)	5 days
2. System Design	Architecture design, database schema (14-04-2025 to 18-04-2025)	5 days
3. Development	Frontend, backend, APIs (21-04-2025 to 30-05-2025)	30 days
4. Testing	Unit, integration, and UAT (02-06-2025 to 06-06-2025)	5 days
5. Deployment	Cloud setup, go-live (10-06-2025 to 14-06-2025)	4 days
6. Training & Handover	User training, documentation (concurrent with deployment) (15-06-2025 to 15-06-2025)	1 days